# Truth as a Compression Artifact in Language Model Training

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## Abstract

When transformers train on contradictory data—the same problem with both correct and incorrect solutions—which answer do they prefer? We hypothesize that next-token prediction, as a compression process, favors whichever answer cluster has lower description length; truth benefits only when errors lack internal structure. We test this by training transformers (3.5M–1B parameters) from scratch on controlled corpora, systematically varying the structure of errors. We find that (a) when errors are random, models develop a correctness preference scaling from 65% to 85% with model size; (b) when errors follow a single coherent alternative rule, this preference vanishes ($\sim$45–51%); (c) two competing wrong rules suffice to restore it (47% $\rightarrow$ 78%). The pattern reproduces on Wikipedia paragraphs with entity substitution (71% vs 46%) and at 1B scale on a mixed natural-text corpus (77% vs 47%). These results are consistent with the hypothesis that, in controlled contradictory corpora, model preference tracks the relative compressibility of competing answer systems rather than truth per se.

## 1 Introduction

Real-world training corpora are noisy: the same question may receive contradictory answers across different documents. Yet language models trained on such data tend to prefer correct information. If we do not understand *why* this filtering works, we cannot predict *when* it will fail—for instance, when errors are not diverse but internally consistent.

Several explanations have been offered: scaling improves factual performance [Kadavath et al., 2022]; frequency matters, since factual accuracy correlates with how often correct information appears in training data [Elazar et al., 2022, Joshi et al., 2024, Kandpal et al., 2023]; and truth-correlated internal representations have been found [Burns et al., 2023, Marks and Tegmark, 2024, Bürger et al., 2024]. Kalai and Vempala [2024] established that calibrated models must hallucinate at a rate bounded by the corpus's monofact rate. Yet none of these explain why the *training objective alone* would favor one answer over another when both appear at equal frequency in the same format.

We propose that the answer lies in the structure of errors. Minimizing cross-entropy is equivalent to minimizing code length [Shannon, 1948, Delétang et al., 2024], linking LLM training to the Minimum Description Length principle [Rissanen, 1978, Grünwald, 2007]. In our task families, a correct rule system is compact; diverse random errors must be memorized individually. But a *coherent* false system—internally consistent, just wrong—can be equally compact, and the preference vanishes.

We test this hypothesis in a controlled experimental setting. We train transformers (3.5M–1B parameters) from scratch on corpora where each mathematical problem appears with both correct and incorrect solutions, systematically varying the structure of errors. This design isolates error structure as the independent variable while holding frequency, format, and domain constant.

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Our contribution is a single experimentally supported hypothesis: **in controlled contradictory corpora, the compression objective favors consistency, not truth.** While the directional prediction (compressible $>$ incompressible) follows from MDL, several quantitative findings do not: the sharpness of the $N{=}1 \to 2$ transition (47% $\to$ 78%), below-chance accuracy for coherent errors ($\sim$45%, not 50%), and inverse scaling of cross-domain verification (accuracy *drops* with model size). We show that (a) random errors produce a correctness preference scaling from 65% to 85%; (b) a single coherent alternative rule eliminates this preference; (c) two competing rules restore it, pinpointing the compressibility boundary; (d) the same contrast reproduces on Wikipedia paragraphs with entity substitution; and (e) the effect persists at 1B parameters on a mixed natural-text corpus (Section 4.7). Whether this extends to large-scale pretraining remains open (Section 5).

## 2 Related Work

**Prediction as compression.** The link between prediction and compression traces to Shannon [1948] and was formalized by Solomonoff [1964] and Rissanen's MDL principle [Rissanen, 1978, Grünwald, 2007]. Delétang et al. [2024] showed that language models function as strong general-purpose compressors. Huang et al. [2024] discovered a linear correlation ($r \approx -0.93$) between compression quality and benchmark performance. Wan and Mei [2025] formalize an argument that LLM training approximates Solomonoff induction. Pan et al. [2025] linked compression to knowledge acquisition and scaling; Chlon et al. [2025] linked compression failures to hallucination. Contemporaneous arXiv preprints by Marty et al. [2026] formalize simplicity bias through MDL, and Aksenov et al. [2026] connect compression to mathematical modeling. We build on the MDL framework by experimentally varying the description length of false answer systems.

**Internal representations and world models.** Compression can give rise to structured internal representations: Li et al. [2023a] found board representations in an Othello model, Gurnee and Tegmark [2024] discovered linear space-time representations in Llama-2, and several studies found truth-correlated representations [Marks and Tegmark, 2024, Burns et al., 2023, Li et al., 2023b, Azaria and Mitchell, 2023, Ravfogel et al., 2025]. Our work operates at the behavioral level: we identify data-level conditions under which compression produces a preference for correct completions, leaving activation-level analysis for future work.

**Truthfulness and data statistics.** Joshi et al. [2024] linked truthfulness to "personas" in pretraining data. Elazar et al. [2022] demonstrated frequency dependence. Kandpal et al. [2023] showed a direct relationship between document count and accuracy. Li et al. [2024] studied LLM preferences under conflicting knowledge and found that formality and surface quality predict which version the model favors—a consistency-driven explanation complementary to ours. A growing body of work studies knowledge conflicts directly: Xie et al. [2024] surveyed approaches to resolving conflicts between parametric and contextual knowledge, and Longpre et al. [2021] showed that entity-substituted evidence can override parametric memory. Unlike these analyses, we fix frequency, format, and surface quality, varying only the compressibility of the error system itself—isolating error structure as the causal variable rather than studying post-hoc conflict resolution.

**Simplicity bias and noisy labels.** Neural networks prefer simple functions [Valle-Pérez et al., 2019, Mingard et al., 2021, Goldblum et al., 2024]. The noisy labels literature directly parallels our setup: Zhang et al. [2017] showed memorization of random labels; Rolnick et al. [2017] showed robustness to massive label noise. Grokking connects to compression as a memorization-to-generalization transition [Nanda et al., 2023, Liu et al., 2023]. Our experiments extend this line by showing that "structured noise"—coherent errors—is not filtered out. To our knowledge, systematic variation of error compressibility in a denoising setting has not been studied before.

## 3 Methodology

**Overview.** We test the hypothesis that next-token prediction favors whichever answer system has lower description length, regardless of its truth value. To isolate this, we train models from scratch on controlled corpora where each problem appears with contradictory answers, varying only the structure of errors. The core experiments compare random errors (high description length) against coherent errors (low description length) at equal frequency (Section 4.1), then probe the boundary with multi-rule errors of intermediate compressibility (Section 4.2). Transfer to natural language is tested via Wikipedia entity substitution (Section 4.3).

### 3.1 Hypothesis and MDL Framing

Consider a corpus where each problem appears with a correct answer (theory $T_1$) and an alternative ($T_2$). An MDL learner [Rissanen, 1978, Grünwald, 2007] minimizes $L(M) + L(D|M)$. The model must encode *both* answer clusters as a latent-source mixture ($T_1$ and $T_2$ with per-problem source assignments). When $K(T_2) \gg K(T_1)$ (random errors), encoding the false cluster requires memorizing individual answers, so the model allocates more capacity to the compressible correct cluster—yielding lower NLL on correct completions. When $K(T_2) \approx K(T_1)$ (coherent errors), both clusters are equally compressible and the model has no basis to prefer one over the other. With $N$ competing false rules, a random "selector" (which rule applies to which problem) adds $\log N$ bits per problem, progressively degrading the false cluster's compressibility (formal details in Appendix B).

Throughout, "truth" means correctness of mathematical derivations and factual accuracy of Wikipedia paragraphs—models compress text, not reality. The compressibility gap is domain-dependent and smaller in natural language than in formal math (Section 4.3). Additional limitations are discussed in the Limitations section.

### 3.2 Models and Training

We train decoder-only transformers with a GPT-2-like design [Radford et al., 2019]: pre-norm, GELU activation, and a causal attention mask. This is a minimal, well-characterized baseline that isolates training-data structure from architectural confounds; Section 4.6 verifies the effect on the more recent Qwen3 architecture [Yang et al., 2025]. Size labels below are internal designations and do not correspond to OpenAI's GPT-2 checkpoints (e.g., our "large" at 86M is smaller than GPT-2 Small at 117M).

| Config | Layers | $d_{\text{model}}$ | Heads | Parameters |
|---|---|---|---|---|
| tiny | 4 | 256 | 4 | $\sim$3.5M |
| small | 6 | 384 | 6 | $\sim$12M |
| medium | 8 | 512 | 8 | $\sim$26M |
| large | 12 | 768 | 12 | $\sim$86M |

Table 1: Model configurations used in all experiments.

Optimizer: AdamW (weight_decay=0.01), cosine decay with linear warmup, lr $=$ 3e-4, seq_len $=$ 256, batch_size $=$ 32, 5000 steps. All experiments use 4 random seeds for core conditions (2 where noted). Learning curves confirm behavioral plateau by step 3000–4000 at all sizes (Appendix F). Denoising experiments use PyTorch 2.3 [Paszke et al., 2019]; multi-rule and Wikipedia experiments use MLX 0.31 [Hannun et al., 2023]. Problems are generated and verified with SymPy 1.14 [Meurer et al., 2017]; no human annotation is involved. Full generation templates are in the supplementary material.

### 3.3 Corpus Design

**Denoising setup (primary).** A generator creates mathematical problems of four types: multi-step arithmetic, factoring, equation solving, and differentiation, formatted as step-by-step derivations in English. The key design: **each problem appears with contradictory answers**—modeling the scenario where the same question receives conflicting responses. The tokenizer is character-level (vocab $\sim$57); BPE robustness is verified in Section 4.4.

**Wikipedia setup (transfer).** To test generalization beyond formal math, we construct corpora from 20,000 English-Wikipedia paragraphs sampled from the 2023-11-01 Wikimedia dump [Wikimedia Foundation, 2023]. Using spaCy 3.8 with the `en_core_web_sm` model for named-entity recognition [Honnibal et al., 2020], we create two corruption modes: *random substitution* (each entity replaced with a random entity of the same type) and *coherent substitution* (a consistent global

| Condition | Correct | Incorrect | Ratio | Purpose |
|---|---|---|---|---|
| Equal random | 1 | 1 random | 1:1 | Signal extraction from random noise |
| Equal coherent | 1 | 1 coherent | 1:1 | Control: coherent errors eliminate bias |
| 2:1 noise | 1 | 2 random | 1:2 | Noise tolerance at moderate noise |
| 4:1 noise | 1 | 4 random | 1:4 | Noise tolerance at high noise |

Table 2: Denoising corpus conditions. Each corpus contains 5,000 unique problems.

| Size | Params | Random Accuracy | Coherent Accuracy | Random Seeds | Coherent Seeds |
|---|---|---|---|---|---|
| tiny | 3.5M | $\mathbf{65.3\% \pm 1.3\%}$ | $43.5\% \pm 2.6\%$ | 4 | 4 |
| small | 12M | $\mathbf{74.6\% \pm 1.6\%}$ | $44.5\% \pm 3.0\%$ | 4 | 4 |
| medium | 26M | $\mathbf{81.1\% \pm 1.2\%}$ | $45.8\% \pm 3.4\%$ | 4 | 4 |
| large | 86M | $\mathbf{85.2\% \pm 2.3\%}$ | $51.0\% \pm 0.8\%$ | 2 | 2 |

Table 3: Denoising paired evaluation: equal random vs equal coherent errors, across model sizes. All $\pm$ values are standard deviations across random seeds. Random accuracy scales with model size; coherent accuracy remains near chance.

mapping, e.g., every "France" $\rightarrow$ "Japan", every "Paris" $\rightarrow$ "Tokyo"). Coherent substitution preserves grammatical structure and cross-entity consistency within each paragraph; random substitution breaks co-occurrence patterns (e.g., a paragraph may mention "Kumamoto" in a context about French history). Both modes substitute entities of matching type, so surface-level grammatical anomalies (gender, case agreement) are minimal. Models are trained on 50/50 mixes; evaluation uses 2,000 held-out paired paragraphs.

### 3.4 Evaluation

**Paired evaluation (primary).** For each problem, a shared prompt is generated with two completions (correct and incorrect). NLL is computed on completion tokens only, conditioned on the prompt. The primary metric is **pair accuracy**: the fraction of pairs where the model assigns lower NLL to the correct completion. This is equivalent to the Common Language Effect Size (CLES; McGraw and Wong, 1992). Length-matched evaluation confirms that accuracy is not driven by completion length differences (within 1 pp across all conditions).

**Generative evaluation.** Greedy decoding on 500 test prompts with automated SymPy verification. This confirms that the discriminative effect extends to generation (Section 4.5; Table 6).

## 4 Results

### 4.1 The Central Contrast: Random vs Coherent Errors

When the same problem appears with both a correct and a random wrong answer, the model learns to prefer the correct one—accuracy reaches 85% at 86M parameters (Table 3, Figure 1). When the wrong answer follows a coherent rule system, the effect disappears: accuracy hovers near 50% at all scales.

Random accuracy increases monotonically with scale ($65\% \rightarrow 75\% \rightarrow 81\% \rightarrow 85\%$). Coherent accuracy remains near chance across all scales, consistent with the MDL prediction that equally compressible systems at equal frequency should be indistinguishable. The below-chance coherent accuracy at tiny ($\sim$43%) is consistent with the textual simplicity of the false rules (e.g., dropping terms in derivatives); the compressor may prefer whichever system is simpler to encode. The multi-rule experiment (Section 4.2) controls for this surface-form effect.

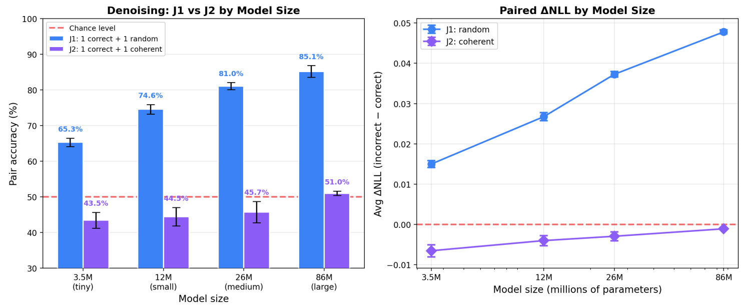


Figure 1: The central contrast. Random errors: accuracy scales from 65% to 85%. Coherent errors: accuracy stays near chance across all sizes. Accuracy = fraction of pairs where the model assigns lower NLL to the correct completion.

| $N$ Rules | Accuracy | $p$ |
|---|---|---|
| 1 (coherent) | $46.6\% \pm 2.4\%$ | n.s. |
| 2 | $77.6\% \pm 1.3\%$ | $< 10^{-6}$ |
| 3 | $82.8\% \pm 0.1\%$ | $< 10^{-6}$ |
| 5 | $84.8\% \pm 0.5\%$ | $< 10^{-6}$ |
| 10 | $88.3\% \pm 0.7\%$ | $< 10^{-6}$ |

Table 4: Multi-rule paired evaluation (tiny, 3.5M, standard 50/50, 4 seeds). The $N$=1 baseline (46.6%) differs from the denoising coherent accuracy in Table 3 (43.5%) because multi-rule experiments use the standard (non-denoising) setup. $N$=2 on small (12M) yields $86.3\% \pm 0.8\%$, confirming the effect strengthens with capacity.

### 4.2 Multi-Rule Errors: The Sharp Crossover

The denoising experiments establish two poles: one coherent rule yields chance, random errors yield strong bias. We probe the boundary by training on corpora with $N$ alternative wrong rules per task type, where each problem's rule is chosen at random. Each rule is compact, but the mapping "problem $\to$ rule" is unpredictable.

This is the most mechanistically revealing result (Table 4, Figure 2). With one false rule, $K(T_2) \approx K(T_1)$. With two rules, the learner must encode a random selector—which rule applies to which problem—adding high-complexity bits that break compressibility. At $N$=10, accuracy (88.3%) exceeds even the random baseline (79.5%), because multi-rule errors combine compactness of individual rules with incompressibility of the selector.

### 4.3 Transfer to Wikipedia Text with Entity Substitution

The random/coherent contrast reproduces (Table 5): random substitution yields 70–71% accuracy (all $p < 10^{-6}$), coherent substitution yields 46–49% (at or below chance). Note that entity substitution tests co-occurrence compressibility (e.g., "Kumamoto" in a French-history context breaks local patterns), not world-knowledge truth in the philosophical sense. The effect is weaker than in math (71% vs 85%) because natural language provides more textual flexibility for locally fluent errors. Unlike math, Wikipedia accuracy saturates near 70% across the tested range—the compressibility gap in natural language does not substantially benefit from additional capacity at this scale. Per-entity-type accuracy ranges from 82% (LOC) to 61% (CARDINAL), with geographic entities showing the strongest effect (Appendix E).

### 4.4 Robustness Checks

**Tokenization.** The primary experiments use a character-level tokenizer (vocab ∼57). To verify that the effect is not an artifact of tokenization, we repeat the core conditions with a BPE tokenizer (SentencePiece, vocab=1000). The effect survives and strengthens: random accuracy increases from

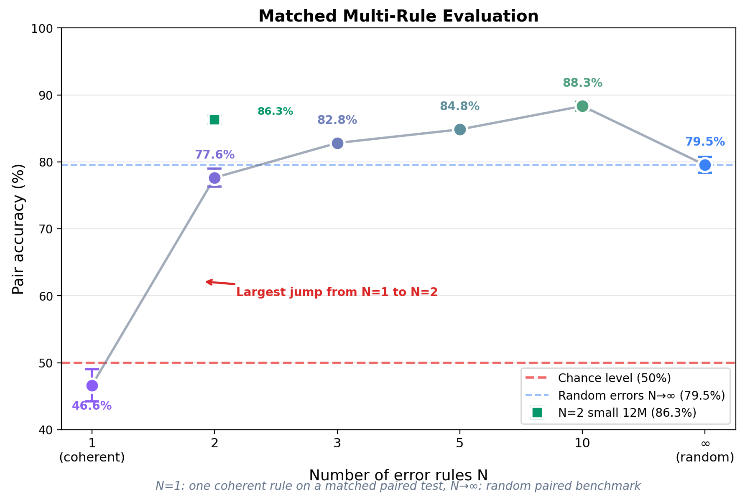


Figure 2: Multi-rule crossover. Accuracy jumps from 47% ($N$=1) to 78% ($N$=2), then grows gradually through $N$=10. X-axis is categorical (not log-scale); point spacing does not represent proportional distances.

| Size | Random Accuracy | Coherent Accuracy |
|---|---|---|
| tiny (3.5M) | $\mathbf{69.6\% \pm 0.1\%}$ | $48.7\% \pm 0.5\%$ |
| small (12M) | $\mathbf{70.7\% \pm 0.4\%}$ | $46.6\% \pm 0.4\%$ |
| medium (26M) | $\mathbf{71.5\% \pm 0.8\%}$ | $46.4\% \pm 0.8\%$ |
| large (86M) | $\mathbf{71.4\% \pm 0.7\%}$ | $45.9\% \pm 1.5\%$ |

Table 5: Wikipedia entity substitution (50/50, 4 seeds per size). The random/coherent contrast reproduces on real text.

65.3% to 75.9% in denoising (79.5% to 85.6% in standard); coherent accuracy remains at chance (49.3% and 45.9% respectively). Full results in Appendix F.

**Standard (non-denoising) setup.** In the standard setup, each problem appears once with either a correct or incorrect solution (no within-problem contradiction). Paired accuracy is higher (79.5–88.3% vs 65.3–85.2%), but the gap closes with scale (from 14 pp at tiny to 3 pp at large). Critically, the random/coherent contrast holds identically: standard coherent accuracy is 47–52%, at chance. Full comparison in Appendix D.

**Frequency vs compressibility.** For random errors, truth bias persists even when correct examples are a small minority: 67% paired accuracy at 10/90 correct-to-incorrect ratio (Appendix D). For coherent errors, the pattern reverses—frequency dominates preference far more strongly: at 40/60 coherent, the model follows the majority with 72% preference for the false system; at 20/80, this rises to 91%. This asymmetry is a direct prediction of the MDL framework: when both systems compress equally well, frequency is the only tiebreaker.

**Matched-control ablation.** A potential confound is that coherent errors affect more derivation steps per problem ($\sim$3 on average) than standard random errors (exactly 1). To control for this, we generate "matched random" errors that corrupt the same number of steps as coherent errors but at random positions with random values. Matched-random accuracy is $74.5\% \pm 0.4\%$ (4 seeds)—lower than standard random (79.5%, which corrupts only 1 step) but far above coherent (47.2%, which also corrupts $\sim$3 steps). The 27 pp gap between matched-random and coherent at identical step counts confirms that compressibility, not surface-level corruption load, drives the random/coherent contrast.

**Noise tolerance.** The correct signal persists under increasing noise ratios (1:2 and 1:4 random), with accuracy degrading gracefully from 85% to 66–75% at large scale. Details in Appendix D.

| Size | Random-trained | Coherent-trained | Gap |
|---|---|---|---|
| tiny (3.5M) | $30.5\% \pm 1.7\%$ | $20.8\% \pm 3.6\%$ | $+9.7$ pp |
| small (12M) | $46.7\% \pm 1.9\%$ | $31.7\% \pm 1.8\%$ | $+15.0$ pp |
| medium (26M) | $50.5\% \pm 1.3\%$ | $36.1\% \pm 2.7\%$ | $+14.4$ pp |
| large (86M) | $52.8\% \pm 1.1\%$ | $35.6\% \pm 1.8\%$ | $+17.2$ pp |

Table 6: Generative accuracy (4 seeds, 500 problems each). The gap widens with model size.

| Condition | seed43 | seed44 | seed45 | Mean |
|---|---|---|---|---|
| Random | 85.3% | 85.7% | 89.3% | **86.8%** |
| Coherent | 49.3% | 52.8% | 49.8% | **50.6%** |

Table 7: Qwen3-0.6B paired evaluation (50/50, 3 seeds). The random/coherent contrast reproduces across architectures.

### 4.5 Generative Evaluation

Paired evaluation is a forced-choice (discriminative) setting. To verify that the effect extends to generation, we run greedy decoding with SymPy verification on 500 problems per model across all sizes.

The gap widens with model size (10 pp at tiny $\rightarrow$ 17 pp at large), confirming that the discriminative preference translates into a generative advantage (Table 6). The absolute generation accuracy is substantially lower than paired accuracy (53% vs 85% at large). This gap reflects the fundamental difference between discrimination and generation: the model may assign higher likelihood to correct completions while lacking the capacity to produce them reliably from scratch. Both metrics agree on the direction: random-trained models outperform coherent-trained ones at all scales.

Per-task breakdown reveals that generation difficulty varies sharply by type. For random-trained large models: derivative (79.5%), algebra (70.4%), equation (61.0%), arithmetic (0%). Multi-step arithmetic requires exact carry propagation across many digits—a well-known difficulty for autoregressive generation at this scale—even though paired evaluation shows strong discrimination. Coherent-trained models show lower generation accuracy across all types, with the gap largest for algebra (70.4% vs 38.3%) and smallest for derivative (79.5% vs 56.5%).

### 4.6 Architecture Robustness: Qwen3

To verify that the effect is not specific to the GPT-2 architecture, we train a Qwen3-0.6B model [Yang et al., 2025] (420M non-embedding parameters, 28 layers, RoPE + GQA + SwiGLU + RMSNorm) from scratch on the same denoising corpora. This architecture differs from GPT-2 in positional encoding, attention mechanism, activation function, and normalization—testing whether the compression-consistency effect is a general property of autoregressive transformers.

The pattern reproduces (Table 7, Figure 3): random errors yield 86.8% accuracy, coherent errors yield 50.6% (chance). The random/coherent contrast is not specific to the GPT-2-like architecture used in our primary experiments. Qwen3's random accuracy (86.8%) is comparable to GPT-2 large (85.2%) despite training on the same corpus sized for smaller models, suggesting further gains with compute-matched training.

### 4.7 Scaling to 1B on Real Text

The preceding experiments use synthetic math corpora and models up to 86M parameters (420M for architecture robustness). A natural objection is that the effect may be specific to toy-scale models or synthetic data. To address both concerns simultaneously, we train a Qwen3-architecture model ($\sim$1B parameters) from scratch on 1 GB of FineWeb-Edu [Penedo et al., 2024]—a curated subset of real web text—with mathematical content filtered out and replaced by 8% contradictory math problems (denoising format, equal random or coherent). This setup tests whether the compression-consistency effect survives in a regime where (a) the model has an order of magnitude more parameters than our

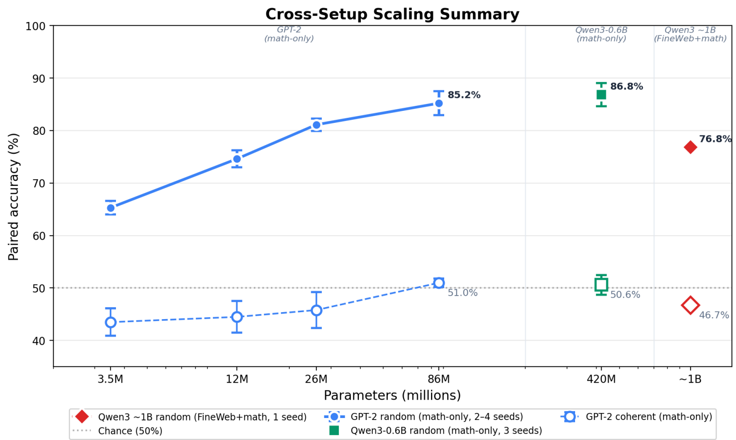


Figure 3: Cross-setup scaling summary. GPT-2 family (3.5M–86M, math-only, 2–4 seeds), Qwen3-0.6B (420M, math-only, 3 seeds), and Qwen3-architecture ∼1B (FineWeb-Edu + 8% math, single seed). Note: setups differ in architecture, corpus, and seed count; this figure summarizes the directional trend, not a single scaling law.

| Condition | Accuracy | ΔLoss | $p^*$ | Seeds |
|---|---|---|---|---|
| Random | **76.8%** | $+0.098$ | $< 10^{-6}$ | 1 |
| Coherent | 46.7% | $-0.003$ | n.s. | 1 |

Table 8: Qwen3-architecture ∼1B trained on FineWeb-Edu + 8% contradictory math (denoising, 50/50). Single-seed sanity check: the random/coherent contrast reproduces at 1B scale on real text. *Per-example Wilcoxon test (single training run); descriptive only—no across-seed uncertainty is available.

largest primary experiment (∼1B vs 86M), and (b) the contradictory signal is a small fraction of a large, naturalistic corpus.

The result (Table 8) confirms that the random/coherent contrast holds: random errors yield 76.8% paired accuracy ($\Delta$Loss $= +0.098$), while coherent errors yield 46.7% (chance). As a single-seed experiment, these results are descriptive; the $p$-value reflects per-example significance within one training run and should not be interpreted as across-run reliability. The random accuracy is lower than the math-only GPT-2 large (85.2%) and Qwen3-0.6B (86.8%), likely because the contradictory math signal constitutes only 8% of the training corpus, diluted by general text. The key finding is that even in this diluted, naturalistic setting, the compression objective still distinguishes random from coherent errors—and the $\Delta$Loss magnitude ($+0.098$) is the largest we observe, suggesting that the 1B model develops a stronger per-pair preference despite lower aggregate accuracy.

## 5 Discussion

**Summary of findings.** The experiments support a unified hypothesis: **in our controlled settings, the compression objective tracks consistency rather than truth.** Truth bias emerges only when false alternatives are structurally incoherent; the evidence converges across setups (denoising, standard, Wikipedia), error types, and evaluation modes (discriminative and generative).

**The role of frequency.** For random errors, compressibility overrides frequency: truth bias persists even at 10/90 correct-to-incorrect ratio (67% accuracy). For coherent errors, frequency dominates: at 40/60 coherent mixing the model prefers the false system at 72%, rising to 91% at 20/80, consistent with the MDL prediction that when systems compress equally, the more frequent one wins. Coherent falsehood therefore needs no frequency advantage to neutralize truth bias.

**What the multi-rule experiment shows.** The $N$=1 → 2 transition provides strong evidence that compressibility, not surface-form complexity, is the operative variable: each rule at $N$=2 remains compact, yet the random rule-to-problem assignment introduces incompressible bits. A probability-mass-splitting alternative (correct gets $\sim 50\%$, each wrong gets $\sim 50\%/N$) would predict $N{=}2$ accuracy approaching 100%; the observed 78% suggests partial, not complete, rule separation, consistent with a compressibility-driven account where the model preferentially allocates capacity to the more compressible cluster.

**Implications and scope.** In our controlled settings, the training objective does not provide an inherent "truth compass"—internally coherent falsehood remains competitive. Cross-domain verification can partially restore truth bias but exhibits inverse scaling: verification accuracy drops from 71% at 3.5M to 61% at 86M parameters (Appendix F; 10 models), as larger models absorb coherent within-domain patterns more readily. This is potentially relevant for alignment (systematic false beliefs may resist filtering at scale), hallucinations (coherent misconceptions may persist; cf. Kalai and Vempala, 2024), and data curation (coordinated errors may evade diversity-based filtering). The 1B experiment on real web text (Section 4.7) extends the effect beyond toy scale and synthetic data, but all our experiments remain far below production LLMs. Whether these findings transfer to large-scale pretraining remains open; we identify a mechanism and its boundary conditions, not direct claims about production models.

## 6 Conclusion

In controlled experiments with transformers from 3.5M to $\sim$1B parameters, models prefer the correct answer only when errors are structurally incoherent: a single coherent alternative rule eliminates this preference, and two competing rules restore it. The pattern reproduces on Wikipedia paragraphs with entity substitution, in generative evaluation, and across architectures. Whether the mechanism extends to large-scale pretraining remains open; in the settings we study, correctness preference is a compression artifact that tracks the description length of the false answer system.

## Limitations

**Model scale.** Primary experiments use 3.5M–86M parameter models; Section 4.7 extends to $\sim$1B parameters on natural text with a single seed per condition. This is far below production LLMs (7B–400B+); multi-billion-scale replication with compute-matched training and multiple seeds remains an important next step.

**Domain specificity.** Mathematics provides an unusually crisp correct/incorrect distinction. The effect weakens in natural language (71% vs 85%), where errors can remain locally fluent. Extending to domains with competing real-world knowledge systems (e.g., conflicting news sources, historical revisionism) remains open.

**Discriminative vs generative gap.** Paired accuracy (85%) substantially exceeds generative accuracy (53%) at large scale: generation requires autoregressive decoding (errors compound), while paired evaluation only ranks likelihoods. The gap narrows with scale (49 pp at tiny, 32 pp at large), and both metrics agree on the random/coherent contrast at all tested scales. Paired accuracy should therefore be read as a controlled diagnostic of model preference, not as a direct measure of output truthfulness.

**Causal identification.** Random and coherent corruptions may differ along axes beyond compressibility (affected steps, surface simplicity). The multi-rule experiment holds individual rule complexity constant but leaves open a probability-mass-splitting account (discussed in Section 5); a predictable-selector variant (deterministic rule-to-problem mapping) and a fully matched control (corruption load, lexical diversity, length) would further strengthen the causal claim.

**Future directions.** Linear probing for "truth" vs "coherence" directions in models trained under our conditions could link behavior to internal representations; interactions with RLHF and in-context learning are natural extensions.

**Ethical considerations.** Internally consistent misinformation can be harder for models to filter than diverse errors; this is a controlled diagnostic finding, not evidence of production-LLM vulnerability. Transparent reporting of such failure modes can inform defensive data curation and evaluation.

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# A Reproducibility

All code, data generation scripts, and evaluation scripts are available at [anonymous repository, included as supplementary material]. Denoising experiments were run on Modal.com T4 GPUs using PyTorch. Standard math and Wikipedia experiments were run on Apple Mac M4 (36GB) using MLX (v0.31.0). Qwen3-0.6B experiments were run on Modal.com A10G GPUs; the $\sim$1B Qwen3-architecture experiments were run on Modal.com A100 GPUs. Over 210 models were trained across all conditions; total compute approximately 150 GPU-hours.

**Corpus generation examples.** Each problem is generated programmatically and verified by SymPy. Below is a representative training example (denoising, equal random condition):

```
Problem: Expand and simplify 3 * (2x + 5) - 4x
Step 1: 3 * (2x + 5) = 6x + 15
Step 2: 6x + 15 - 4x = 2x + 15
Answer: 2x + 15
```

The corresponding random-error version replaces a derivation step with a plausible but incorrect computation (e.g., `6x + 15 - 4x = 2x + 11`). In the coherent condition, a systematic rule is applied consistently across all problems of the same type (e.g., for distribution: $a(b + c) = ab + c$ instead of $ab + ac$). The paired test set presents the same prompt with both correct and incorrect completions for NLL comparison.

# B Formal MDL Predictions

We state the theoretical predictions that motivate our experiments. All three describe the behavior of an idealized two-part MDL learner [Rissanen, 1978, Grünwald, 2007], not formal theorems about neural network training dynamics. Our experiments (Sections 4.1–4.3) test whether gradient-descent-trained transformers approximate this behavior.

**Setup.** Consider a corpus $D$ containing $n$ problems. In the denoising setup, each problem $i$ appears with two answers: one generated by a correct rule system $T_1$ and one by an alternative system $T_2$. The model must encode *both* answers—it cannot simply "select" one theory and discard the other. We model this as a latent-source mixture: for each problem, a latent variable $z_i \in \{1, 2\}$ indicates which source generated the answer. The model encodes: (1) the mixture structure (source theories + mixing weights), and (2) the data given the mixture, including the per-problem source assignments.

The total description length is:

$$L(M) + L(D|M) = \underbrace{L(T_1) + L(T_2)}_{\text{theories}} + \underbrace{L(\{z_i\})}_{\text{source assignments}} + \underbrace{\sum_i L(a_i|T_{z_i})}_{\text{answers given sources}}$$

The key quantity is the cost of encoding the alternative system $T_2$ plus the residuals that $T_2$ cannot predict. A model that internalizes the more compressible theory can encode answers from that theory cheaply and must pay a higher per-token cost for answers from the less compressible theory—which is exactly what paired evaluation measures (lower NLL on the more compressible cluster).

**Prediction 1** (random errors). *If $T_1$ is a compact rule system with $L(T_1) = O(1)$ and $T_2$ consists of independent random answers, then correct-answer completions receive lower NLL than incorrect ones.*

*Proof sketch.* The correct answers are generated by a compact $T_1$: once internalized ($O(1)$ bits), each correct answer costs $O(c)$ bits to encode. The random false answers have no shared structure: encoding each requires memorizing an independent value, contributing $O(n)$ total bits to $L(T_2)$.

The model's optimal strategy allocates more probability mass to the predictable (correct) cluster, resulting in lower NLL for correct completions.

**Prediction 2** (coherent errors). *If both $T_1$ and $T_2$ are compact rule systems with $L(T_1) \approx L(T_2) = O(1)$, then at equal frequency the model assigns comparable NLL to both clusters.*

*Proof sketch.* Both theories have comparable model complexity: $L(T_1) \approx L(T_2)$. Given either theory, the per-answer encoding cost is the same: $L(a_i|T_1) \approx L(a_i|T_2)$ for their respective answers. Since both clusters are equally compressible and equally frequent, the optimal mixture assigns equal probability mass to each, yielding comparable NLL.

**Prediction 3** (multi-rule errors). *If the false system consists of $N$ compact rules $\{r_1, \ldots, r_N\}$ with random assignment to problems, then encoding the false cluster requires an additional $O(n \log N)$ bits for source selection, restoring the NLL advantage of the correct cluster.*

*Proof sketch.* Each rule $r_i$ is compact: $L(r_i) = O(1)$. However, encoding which rule applies to which problem requires a selector $s : \{1, \ldots, n\} \to \{1, \ldots, N\}$. When assignments are random (i.i.d. uniform), the selector has entropy $n \log_2 N$ bits and is incompressible. The total false-cluster code length is $L(\{r_i\}) + L(s) + \sum_i L(a_i|r_{s(i)}) = O(N) + O(n \log N) + O(n \cdot c)$. For $N \geq 2$, the $n \log N$ term makes the false cluster more expensive to encode than the correct cluster, which requires no selector. Note that this prediction concerns the *total* encoding cost of the false cluster, not only the per-completion probability mass split across $N$ rules (see Section 5 for discussion of this alternative explanation).

# C Compression Measure (gzip)

To operationalize the MDL argument, we measure compression ratios (gzip level 9) on concatenated correct vs incorrect completions from each paired test set.

| Condition | Correct | Incorrect | Delta | Paired Accuracy |
|---|---|---|---|---|
| random 50/50 | 0.1627 | 0.1639 | +0.0012 | 79.5% |
| coherent 50/50 | 0.1656 | 0.1658 | +0.0002 | 47.2% |
| contradictory | 0.1745 | 0.1744 | −0.0001 | 49.0% |
| multirule $N$=2 | 0.1671 | 0.1710 | +0.0038 | 77.6% |
| multirule $N$=3 | 0.1669 | 0.1722 | +0.0053 | 82.8% |
| multirule $N$=5 | 0.1672 | 0.1730 | +0.0057 | 84.8% |
| multirule $N$=10 | 0.1676 | 0.1752 | +0.0076 | 88.3% |
| world random | 0.0370 | 0.0516 | +0.0146 | 57.7% |
| world coherent | 0.0374 | 0.0384 | +0.0010 | 46.6% |

Table 9: Compression ratio (gzip) and paired accuracy across all 9 conditions. Spearman $\rho = 0.68$, $p = 0.042$.

Conditions with larger compression gaps produce stronger truth bias. We present this as supporting evidence; the primary argument rests on the experimental contrasts in Section 4.

# D Standard (Non-Denoising) Experiments

## D.1 Standard Corpus Design

In the standard setup, each problem appears once with either a correct or incorrect solution. The two groups do not share prompts.

## D.2 Standard vs Denoising

The denoising setup is harder (within-problem contradiction vs across-corpus mixing), but the gap closes with scale.

| Size | Standard | Denoising | Gap |
|---|---|---|---|
| tiny | 79.5% | 65.3% | −14.2 pp |
| small | 86.2% | 74.6% | −11.6 pp |
| medium | 87.1% | 81.1% | −6.0 pp |
| large | 88.3% | 85.2% | −3.1 pp |

Table 10: Standard vs denoising accuracy (50/50 random). The gap closes with scale.

| Proportion | Random | Coherent | False pref. |
|---|---|---|---|
| 50/50 | 80% | 47.2% | 52.8% |
| 40/60 | 79% | 27.8% | 72.2% |
| 30/70 | 75% | 14.7% | 85.3% |
| 20/80 | 69% | 9.6% | 90.4% |
| 10/90 | 67% | – | – |

Table 11: Random vs coherent accuracy across proportions (standard, tiny). Random: truth bias persists at 10/90. Coherent: the model follows pure frequency. "False pref." = $1 -$coherent accuracy, i.e., the fraction of pairs where the model prefers the coherent false completion.

### D.3 Frequency Effects

### D.4 Noise Tolerance

| Condition | Ratio | Tiny | Small | Medium | Large |
|---|---|---|---|---|---|
| Equal random | 1:1 | 65.3% | 74.6% | 81.1% | 85.2% |
| 2:1 noise | 1:2 | 59.0% | 68.6% | 73.6% | 75.2% |
| 4:1 noise | 1:4 | 56.6% | 64.9% | 66.6% | 65.8% |

Table 12: Paired accuracy at increasing noise ratios (denoising). Accuracy degrades gracefully; at 4:1 noise, it plateaus at medium/large.

### D.5 NLL Distribution

Random errors produce a right-skewed per-pair NLL difference (mean $+0.048$, median $+0.025$, 81.5% positive). Coherent errors produce a symmetric distribution (45.5% positive). This explains why pair accuracy and mean $\Delta$Loss can diverge.

## E Additional Transfer Experiments

### E.1 Synthetic World

A synthetic world with 50 entities, 4 types, 15 deterministic rules. Random 50/50: $57.7\% \pm 1.7\%$. Coherent 50/50: $46.6\% \pm 1.7\%$. Per-type: minerals best (68.7%), potions near chance (49.1%).

### E.2 Wikipedia Per-Entity-Type

### E.3 Cross-Domain Falsification

Adding correct cross-domain tasks to a coherent corpus selectively increases derivative accuracy ($35\% \rightarrow 56\%$ at 25%). The effect is non-monotonic (drops at 50% due to dilution).

| Entity Type | $N$ Pairs | Accuracy |
|---|---|---|
| LOC | 45 | 82.2% |
| NORP | 197 | 78.7% |
| GPE | 301 | 77.1% |
| PERSON | 402 | 69.9% |
| ORG | 664 | 65.2% |
| CARDINAL | 172 | 61.0% |

Table 13: Wikipedia accuracy by entity type (random substitution, averaged across sizes).

| Setup | Tokenizer | Random | Coherent |
|---|---|---|---|
| Standard | Char | 79.5% | 47.2% |
| Standard | BPE | **85.6**% | 45.9% |
| Denoising | Char | 65.3% | 43.5% |
| Denoising | BPE | **75.9**% | 49.3% |

Table 14: BPE vs char-level (tiny, 4 seeds). The effect survives BPE and strengthens for random errors.

# F Robustness Checks

## F.1 BPE Tokenization

## F.2 Chained Verification

Embedding cross-domain verification dependencies in coherent-error tasks restores truth bias but exhibits inverse scaling across 10 models (4 tiny, 4 small, 2 large):

| Size | Paired Accuracy | Seeds | Standard coherent baseline |
|---|---|---|---|
| tiny (3.5M) | $70.9\% \pm 1.2\%$ | 4 | 43.5% |
| small (12M) | $64.2\% \pm 1.5\%$ | 4 | 44.5% |
| large (86M) | $60.6\% \pm 1.2\%$ | 2 | 51.0% |

Table 15: Chained-verification paired accuracy vs the corresponding standard coherent baseline (Table 3). Verification raises accuracy above chance at all sizes, but the advantage shrinks with scale: larger models absorb coherent within-domain patterns more readily, making embedded cross-domain checks less effective.

## F.3 Learning Curves

All sizes reach behavioral plateau by step 3000–4000. The large model achieves 88.8% at step 3000, stable through 5000.

# G Per-Seed Details

Tables 16–17 report per-seed accuracy for the core denoising conditions. Per-seed breakdowns for multi-rule, Wikipedia, and Qwen3 experiments are available in the supplementary code repository (evaluation logs per model).

# NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

| Size | Seed 1 | Seed 2 | Seed 3 | Seed 4 | Mean |
|---|---|---|---|---|---|
| tiny | 64.2% | 64.1% | 67.0% | 65.8% | 65.3% |
| small | 75.1% | 76.3% | 72.6% | 74.2% | 74.6% |
| medium | 79.6% | 80.9% | 82.4% | 81.3% | 81.1% |
| large | 83.5% | 86.8% | – | – | 85.2% |

Table 16: Denoising equal random, per-seed accuracy.

| Size | Seed 1 | Seed 2 | Seed 3 | Seed 4 | Mean |
|---|---|---|---|---|---|
| tiny | 40.2% | 44.3% | 46.3% | 43.0% | 43.5% |
| small | 44.8% | 45.3% | 40.3% | 47.4% | 44.5% |
| medium | 45.8% | 40.9% | 47.7% | 48.6% | 45.8% |
| large | 51.6% | 50.4% | – | – | 51.0% |

Table 17: Denoising equal coherent, per-seed accuracy.

Justification: All claims are explicitly qualified to our controlled experimental setting. The abstract states the scope (3.5M–1B parameters, controlled corpora) and the introduction notes that extension to large-scale pretraining is an open question.

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